

ORDERED RANDOM SELECTION WITH PROBABILITIES PROPORTIONAL TO SIZE

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Summary: Various algorithms exist for obtaining a simple random sample drawn without replacement during a single run through a list of entries in a computer's memory. In audit practice it is sometimes necessary to draw a random sample without replacement and with probabilities proportional to size of each sampling unit. Two algorithms for this purpose are proposed.

1. SIMPLE RANDOM SAMPLING

The need often arises to draw a random sample from units of a population arriving sequentially at the sampler. As each unit arrives a decision is required either to include it in the sample, or to let it pass (Fan, Muller and Rezucha (1962)). This is particularly the case in sampling without replacement from items stored in computer memory. Algorithms have been developed, based on the principle of sequential selection, to draw a simple random sample in a single run through the computer's memory. The following cases may be distinguished if a simple random sample from a single stratum is required:

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1.1 Drawing a sample of *expected size* n from a population of size N using N random numbers where N is known. This is a binomial distribution approach where selections are independent with constant probability (Fan, Muller and Rezucha (1962)).

1.2 Drawing a sample of *exactly* n from a population of N using almost N random numbers with N known (Fan, Muller and Rezucha (1962), Bebbington (1975)).

1.3 Drawing a sample of exactly n from a population of N using *only* n random numbers with N known (Bissell (1986)).

1.4 Drawing a sample of exactly n from a population of N using $N - n$ random numbers with N *unknown* (McLeod and Bellhouse (1983)).

1.5 Drawing a sample of exactly n from a population of N using $2n$ random numbers with N unknown (Pinkham (1987)).

The development in the algorithms for cases 1.1 through 1.5 above is obvious, namely obtaining a sample of specified size, less frequent calls on a random number generator and elimination of the need for the population size to be known beforehand.

2. SAMPLING WITH PROBABILITIES PROPORTIONAL TO SIZE (PPS)

It is well known that by assigning varying probabilities of selection to different units in a population, it is possible to reduce considerably the sampling error of the estimates over those obtained when sampling with equal probabilities (Raj, (1956)). Hanif and Brewer (1980) list 50 selection procedures from the literature by which a PPS sample may be obtained and Chaudhuri and Vos (1988) give an up-to-date review of sampling from finite populations with unequal probabilities. Yates and Grundy (1953) describe what they call "the usual method of selection" as follows: "...select the first unit with probability proportional to size, and the second with probability proportional to size of the remaining units, and so on".

Denote the size of the i -th unit, U_i , in a population of N units by a_i and

let $R_1 = \sum_{i=1}^N a_i$ be the total of the size measures for all the units in the population. Typical size units are the size of a farm, the monetary value of an invoice or the number of people in a household. The usual method of selection of a PPS sample then requires that the probability of including a specific population unit in the sample when n units are selected randomly from N population units should be na_i/R_1 . The use of these inclusion probabilities when sampling without replacement, gives rise to bias in estimation. However, Yates and Grundy (1953) show "that the amount of this bias is usually quite trivial". If this is true, and if a valid algorithm is used to select the sample, then in a simulation of, say, 10 000 samples, the frequency with which the i -th population unit appears in these samples should not differ significantly from $10000(na_i/R_1)$, $i = 1, 2, \dots, N$. This will be tested when using the algorithms for drawing a PPS sample proposed in the later sections of this paper.

Further, for the purpose of testing such algorithms, a population of $N = 50$ units was created for which the unit sizes vary randomly between 25 and 100.

3. THE SIEVE METHOD

Driessen (1988) discusses the so-called "sieve method" of sampling used for audit practice. The units in the population are of different sizes, measured in "dollars". Each unit is considered for inclusion in the sample with probability of its size relative to the total size (dollars) for all units. Driessen (1988, p.121) states that "a practical disadvantage of the sieve method may be that the real sample size, contrary to most sampling methods, is random, just like the number of bad dollars found."

The sieve method is a PPS analog of case 1.1 above for simple random sampling, where the sample size likewise was a random variable. Driessen's statement may be illustrated by a simulation whereby 10 000 samples of intended size 5 were drawn from the population described in the last sentence of section 2 above.

The algorithm for the sieve method may be written as follows:

Assume that the sample size is large enough that $a_i \leq R_1/n$,
then:

3.1 Set $i = 0, R_1 = \sum_{i=1}^N a_i$

3.2 $i \rightarrow i + 1$ and generate z , a uniform random variate on $(0,1)$.

3.3 If $z \leq a_i/n/R_1$ select unit i . Return to step 3.2.

Each unit is selected with inclusion probability proportional to its size and the expected sample size is n (Driessen (1988)).

The following is a distribution of the sample sizes actually obtained in 10 000 samples:

n	0	1	2	3	4	5	6	7	8	9
$f(n)$	47	290	762	1405	1841	1848	1553	1064	624	313
n	10	11	12	13	14					
$f(n)$	149	71	25	5	3					

The range of variation in the sample size is clearly illustrated.

The χ^2 -value for measuring the closeness of fit of the observed frequencies with which the individual members of the population were included in the samples, to the expected frequencies described in section 2 (viz. $10000(na_i/R_1), i = 1, 2, \dots, 50$), was $\chi^2 = 64, 2$, which is not significant. ($\chi^2_{49;0,95} = 66, 3$.)

4. AN EXACT SAMPLE SIZE, n

The question now arises whether an algorithm for PPS sampling can also be developed which will be an analog to the simple random sampling case in 1.2 of section 1. In order to investigate this some more notation is needed.

Let, as in section 2 above, $R_1 = \sum_{i=1}^N a_i$ be the population total, $R_2 = \sum_{i=2}^N a_i$, or $R_2 = R_1 - a_1$ be the population total for the units remaining after unit one had been considered for inclusion, and in general

$$R_k = \sum_{i=k}^n a_i = R_1 - \sum_{i=1}^{k-1} a_i.$$

Let the inclusion probability for the first unit considered be $p_1 = a_1 n / R_1$. Let the conditional inclusion probability for unit two, given that unit one

has not been selected, be proportional to its size relative to the total size of the units not yet considered for inclusion, and with n units still to be selected this may be written as $p_2 = a_2 n / R_2$. Likewise, let the conditional inclusion probability for unit two, given that unit one has been selected without replacement, and $n - 1$ further selections must be made, be $p_2 = a_2(n - 1) / R_2$. Now consider the following algorithm:

- 4.1 Set $i = 0$ and dummy variables $R' = R_1$ and $n' = n$.
- 4.2 $i \rightarrow i + 1$ and generate z , a uniform random variate on $(0,1)$.
- 4.3 If $z \leq a_i n' / R'$ select unit i , reduce n' by one and R' by a_i and return to step 4.2.
- 4.4 If $z > a_i n' / R'$ reduce R' by a_i and return to step 4.2.

Note that, if $n - 1$ units have been selected when the last (N -th) population unit is considered, then $p_N = a_N(n - (n - 1)) / R_N = 1$ because $R_N = a_N$. The N -th unit will thus certainly be selected to make up the sample of n . If, however, $n - 2$ units have been selected when population unit $N - 1$ is considered for inclusion, then

$$\begin{aligned}
 (4.1) \quad p_{N-1} &= \min\{1, a_{N-1}(n - (n - 2)) / R_{N-1}\} \\
 &= \min\{1, 2a_{N-1} / (a_{N-1} + a_N)\} \\
 &< 1 \text{ if } a_{N-1} < a_N.
 \end{aligned}$$

Thus with a small probability unit $N - 1$ will not be selected and the sample size will not reach n .

In general, denote by k^i the number of units already selected for the sample when the i -th population unit in the sequence is considered for inclusion. Let U'_i be the event that unit U_i is not selected. By similar arguments as used above we may write the probability that a sample will contain less than n elements as:

$$\begin{aligned}
 (4.2) \quad P[k^N < n] &= P[U'_{N-n+1} | k^{N-n} = 0] \\
 &\quad + (N - n + 1) \text{ terms of the form}
 \end{aligned}$$

$$\begin{aligned}
 & P[U'_{N-n+2} | k^{N-n+1} = 1] \\
 & + \dots + ({}^N P_{N-1}) \text{ terms of the form} \\
 & P[U'_N | k^{N-1} = n-1]
 \end{aligned}$$

$$\begin{aligned}
 & = P[U'_1 \cap U'_2 \cap \dots \cap U'_{N-n+1}] + \dots \\
 & = \left(1 - \frac{na_1}{R_1}\right) \left(1 - \frac{na_2}{R_2}\right) \dots \left(1 - \frac{na_{N-n+1}}{R_{N-n+1}}\right) + \dots
 \end{aligned}$$

By arguments such as in (4.1) above it follows that, if $a_{N-n+1} \geq a_{N-n+2} \geq \dots \geq a_N$, then

$$\left(1 - \frac{na_{N-n+1}}{R_{N-n+1}}\right) \leq 0$$

and hence the probability of the first term on the right-hand side of (4.2) will be

$$\max \left\{ 0, \left(1 - \frac{na_1}{R_1}\right) \left(1 - \frac{na_2}{R_2}\right) \dots \left(1 - \frac{na_{N-n+1}}{R_{N-n+1}}\right) \right\} = 0$$

This will hold for all terms on the right-hand side of (4.2) under the stated conditions, hence

$$P[k^N < n] = 0 \text{ if } a_{N-n+1} \geq a_{N-n+2} \geq \dots \geq a_N.$$

On the other hand, if $a_{N-n+1} < a_{N-n+2} < \dots < a_N$, then, as in (4.1),

$$1 > \left(1 - \frac{na_{N-n+1}}{R_{N-n+1}}\right) > 0$$

The first term in the right-hand side of (4.2) will thus consist of the product of a sequence of numbers between zero and one, generally decreasing and dependent on the sizes of the population units. It may, however, be expected that for N large relative to n this product will be very small.

For a sample of 5 units drawn from the population described in section 2 above, the probability of not selecting any of the first 46 units when $a_{46} < a_{45} < \dots < a_{50}$, i.e. the first term of (4.2) above, is $p = 0.25 \times 10^{-7}$.

Thus it is clear that the probability of not reaching a sample size n depends on the ordering, and differences in sizes, of successive population units, but can be expected to be low.

To illustrate the efficiency of this algorithm for the population described in section 2 above, 10 000 independent samples of intended size five were drawn from the population. When the last five units in the population were arranged in increasing order of size, three samples each contained four members only while 9 997 samples contained five members each. When the last five units in the population were arranged in decreasing order of size all samples contained five members each.

The χ^2 -value for measuring the closeness of fit of observed to expected frequencies with which individual population units were selected when the population was in a random order, was 60.5, which is not significant.

The above algorithm is a PPS analog of case 1.2 above for simple random sampling.

5. SAVING RANDOM NUMBERS

Following the ideas of Bissell (1986), the probability of not selecting the first m units considered in PPS sampling is given by

$$\frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_m - a_m n}{R_m}$$

If z_1 is a random number on (0,1) the k -th unit will be the first selection if z_1 lies between

$$\frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_{k-1} - a_{k-1} n}{R_{k-1}}$$

and

$$\frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_{k-1} - a_{k-1} n}{R_{k-1}} \frac{R_k - a_k n}{R_k}$$

The probability of this occurrence is

$$\begin{aligned} & \frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_{k-1} - a_{k-1} n}{R_{k-1}} \left(1 - \frac{R_k - a_k n}{R_k} \right) \\ &= \frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_{k-1} - a_{k-1} n}{R_{k-1}} \frac{a_k n}{R_k} \end{aligned}$$

The probability that unit ℓ will be the next to be selected is

$$\frac{R_{k+1} - a_{k+1}(n-1)}{R_{k+1}} \dots \frac{R_{\ell-1} - a_{\ell-1}(n-1)}{R_{\ell-1}} \frac{a_{\ell}(n-1)}{R_{\ell}}$$

and so on.

The product of these probabilities is obviously the probability of obtaining the specific sample containing units $k, \ell, \dots$ etc.

Hence consider the following algorithm:

5.1 Set $R_1 = \sum_{j=1}^N a_j$; n the required sample size and $i = 1$.

5.2 Generate z , a uniform random number on $(0,1)$ and set $p = 1$.

5.3 $p \rightarrow p \left(\frac{R_i - a_i n}{R_i} \right)$.

5.4 If $p \leq z$, select the i -th unit, set $i = i + 1$ and $n = n - 1$. Return to step 5.2.

5.5 If $p > z$ set $i = i + 1$ and return to step 5.3.

Using this algorithm only n random numbers are required to draw a sample of size n . Again, as in (4.2) the probability of not reaching a sample size of n can be written as:

$$\begin{aligned}
 P[k^N < n] &= P[U'_{N-n+1} | k^{N-n} = 0] + \dots \\
 (5.1) \quad &= P \left[\frac{R_1 - a_1 n}{R_1} \frac{R_2 - a_2 n}{R_2} \dots \frac{R_{N-n+1} - a_{N-n+1} n}{R_{N-n+1}} \geq z_1 \right] \\
 &\quad + \dots \\
 &= 0 + \dots \text{ if } R_{N-n+1} \leq a_{N-n+1} n, \\
 &\quad \text{i.e. if } a_{N-n+1} \geq a_{N-n+2} \geq \dots \geq a_N.
 \end{aligned}$$

Under this condition all the terms on the right-hand side of (5.1) will be zero. On the other hand, if $a_{N-n+1} < a_{N-n+2} < \dots < a_N$, the probabilities on the right-hand side of (5.1) may be positive, though they can be expected to be small.

In 10 000 repetitions of drawing a sample of five from the population (in random order) described in section 3, only four were of size four while 9 996 contained five units. The χ^2 -value for measuring the closeness of fit of the observed to expected frequencies with which individual population units

were selected was 46,3; hence it may be concluded that each population unit was indeed selected with probability proportional to its size.

This algorithm is PPS analog to case 1.3 above for simple random sampling.

6. THE POPULATION

For the two algorithms above it was stated that the probability of not reaching a full sample size may be expected to be low, and this was verified by simulation from the small population described in section 2. The question remains whether this could not be different for very skew or very peaked populations. To investigate this the simulations were repeated using both algorithms on nine different populations of 50 units each, each having a beta distribution but ranging from very skew (parameters $p = 0,5, q = 8$) to very peaked ($p = 4, q = 4$) to U-shaped ($p = 0,25, q = 0,25$). The worst result occurred when $p = 4$ and $q = 4$, when 25 samples out of 10 000 were incomplete, though with further repetitions of this case a range of between 1 and 25 incomplete samples were recorded. Otherwise the form of the population did not display any marked influence.

7. CONCLUSION

Two algorithms developed for sampling with probabilities proportional to size yield the exact sample size with a very high probability. The problem of drawing a PPS sample in a single run through a computer's memory when the total of all the size measurements for the whole population, R_1 , is not known in advance (analogous to 1.4 and 1.5 above), remains unsolved.

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